

21. Swap the cases of a specified character column**Code**

```
import pandas as pd

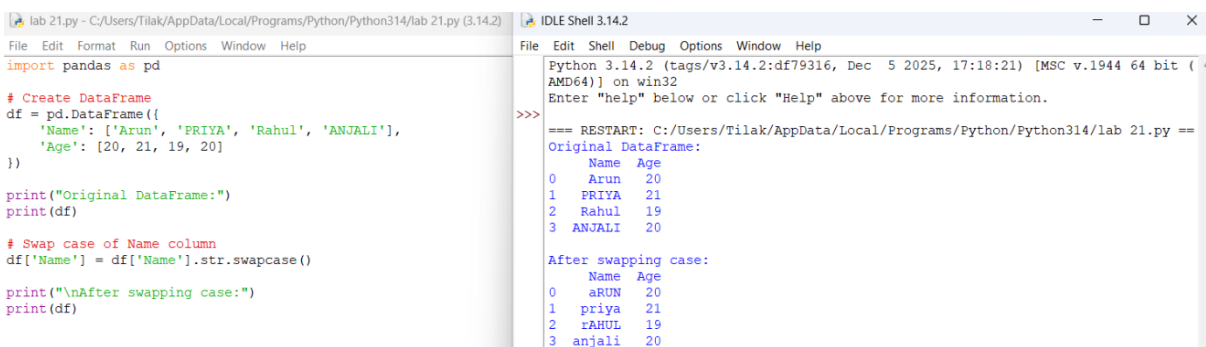
# Create DataFrame

df = pd.DataFrame({
    'Name': ['Arun', 'PRIYA', 'Rahul', 'ANJALI'],
    'Age': [20, 21, 19, 20]
})

print("Original DataFrame:")
print(df)

# Swap case of Name column
df['Name'] = df['Name'].str.swapcase()

print("\nAfter swapping case:")
print(df)
```



The screenshot shows a Python IDE with two windows. The left window displays the code from the previous block. The right window shows the output of the script, which includes the original DataFrame and the DataFrame after swapping the case of the 'Name' column.

```
lab 21.py - C:/Users/Tilak/AppData/Local/Programs/Python/Python314/lab 21.py (3.14.2)
File Edit Format Run Options Window Help
import pandas as pd

# Create DataFrame
df = pd.DataFrame({
    'Name': ['Arun', 'PRIYA', 'Rahul', 'ANJALI'],
    'Age': [20, 21, 19, 20]
})

print("Original DataFrame:")
print(df)

# Swap case of Name column
df['Name'] = df['Name'].str.swapcase()

print("\nAfter swapping case:")
print(df)
```

```
IDLE Shell 3.14.2
File Edit Shell Debug Options Window Help
Python 3.14.2 (tags/v3.14.2:df79316, Dec 5 2025, 17:18:21) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.

>>>
=== RESTART: C:/Users/Tilak/AppData/Local/Programs/Python/Python314/lab 21.py ===
Original DataFrame:
   Name  Age
0  Arun   20
1 PRIYA   21
2  Rahul   19
3 ANJALI   20

After swapping case:
   Name  Age
0 aRUN   20
1 priya  21
2 rAHUL  19
3 anjali  20
```

22. Draw a line with suitable X-axis, Y-axis labels and title

Code

```
import matplotlib.pyplot as plt
```

```
x = [1, 2, 3, 4, 5]
```

```
y = [2, 4, 1, 5, 3]
```

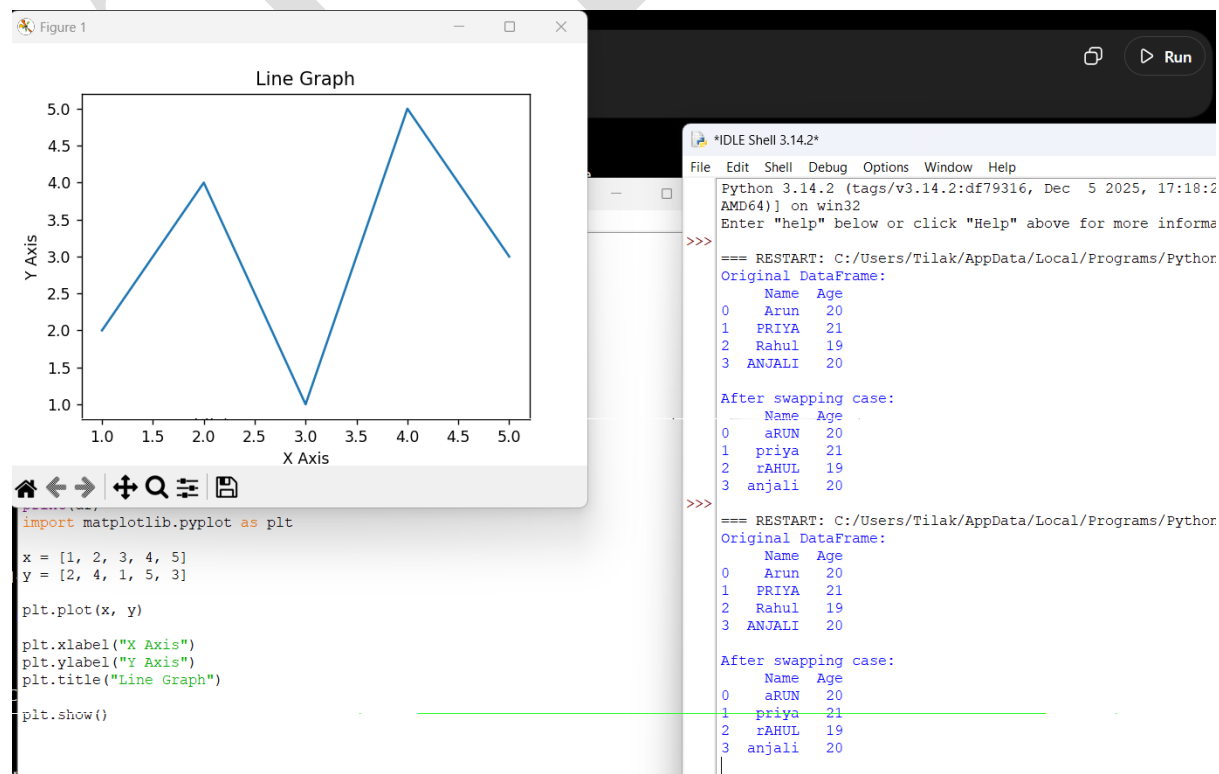
```
plt.plot(x, y)
```

```
plt.xlabel("X Axis")
```

```
plt.ylabel("Y Axis")
```

```
plt.title("Line Graph")
```

```
plt.show()
```



23. Draw a line using values from a text file

First create a text file called:

test.txt

Put exactly:

1 2

2 4

3 1

Save it in the **same folder as your Python program.**

Code

```
import matplotlib.pyplot as plt
```

```
x = []
```

```
y = []
```

```
# Read data from text file
```

```
with open("test.txt", "r") as file:
```

```
    for line in file:
```

```
        values = line.split()
```

```
        x.append(float(values[0]))
```

```
        y.append(float(values[1]))
```

```
# Draw line
```

```
plt.plot(x, y)
```

```
plt.xlabel("X Axis")
```

```
plt.ylabel("Y Axis")
```

```
plt.title("Line Graph from Text File")
```

```
plt.show()
```

Graph values

The file contains:

X	Y
---	---

1	2
---	---

2	4
---	---

3	1
---	---

